

Week 4	
Objectives	We have three main objectives for this week: - An introduction to systems of linear equations (SLE) and their matricial expression - To study and discuss the existence of solutions of a SLE - To try to solve the second questionnaire of Challenge 2.
Session 10	As an introduction, you can watch the second part of the video “Matrices and systems of equations”, where the systems of linear equations are introduced as intersections between planes and lines. You’ll also see how to use the Rouché-Capelli theorem to classify the systems of equations in terms of their solutions, using the concept of rank. Let’s continue with the PDF resources now. Open the Module 2: “Systems of linear equations”, and start reading in section 2, which probably will be familiar to most of you. You’ll see that the objective of this module, at the level of this course, is to look for a method of solving SLE for an indeterminate number of equations and unknowns, as well as their geometric interpretation. Read carefully the first example in the introduction, and try to re-do it. In the second section SLE are defined, and they are classified depending on their number of solutions. It is important to really understand the notation on inconsistent, dependent and independent systems. Follow the resolution of example 1 and explore the possibilities of CalcME for solving systems of equations (linear or not). You can follow the CalcME guide, section “Systems of Linear Equations”. The next step will be to understand the matrix representation of SLE defined in section 3. It is important to understand how the coefficient matrix multiplied by the unknown vector reproduces the original equations of the system. Try to re-do example 2 and write the matricial representation of exercises 2, 3, 4 and 5.
Session 11	Read carefully section 4. Rouché-Capelli theorem states that the only necessary condition to classify a system is a comparison of the rank of the matrix of coefficients with the rank of the bounded matrix. This is just an application of the concept of the rank of a matrix. Do examples 3 and 4. Example 5 is very interesting because it proposes a discussion of a family of systems, depending on the values for certain parameters. We can use the same method as if there weren’t any parameters, but without forgetting to analyze the possible values for them. With this, we can finish exercise 2. Section 5 presents homogeneous SLE. They shouldn’t be difficult. Finish exercise 3.
Session 12	This session doesn’t have any specific program. Today you have a good chance to try doing a video explaining your exercise. This will be a perfect training for when you’ll have to do it for the Synthesis CAT. Try to finish the content of the previous sessions in case you didn’t have time to finish them. If you finished them, continue with the sessions for the next week.
Keep in mind	Remember how important it is to do the exercises yourselves (you can consult the solutions afterwards). These subjects aren’t really difficult, and it will let you really master the methods used here. Some parallel work with CalcME will help you with your study of this module. Remember that you have to solve the second questionnaire of Challenge 2 this week.